QNSMARTI:H:QPCODE

Reg.No : .....

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) QNSMARTI:H:EXTYPE

EXAMINATION QNSMARTI:H:Y:MONTH QNSMARTI:H:Y:YEAR

QNSMARTI:H:S: V SEMESTER

QNSMARTI:H:C:DSE

QNSMARTI:H:C:CODE – MG5DSECHEA01

QNSMARTI:H:C:NAME – FORENSIC CHEMISTRY

(QNSMARTI:H:C:REVYEAR ADMISSION ONWARDS)

Duration: 2hrs **Maximum Marks: 70marks**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer all questions

Each question carries **1** marks

1. Arson refers to:

   a) Natural fire b) Industrial fire c) Intentional setting of fire d) Forest fire

2. Smoke patterns are useful in determining:

   a) Blood group b) Direction of fire spread c) Fingerprints d) DNA profile

3. The term "fatal dose" refers to:

   a) The smallest detectable dose of a poison b) The maximum safe dose of a poison

   c) The amount of poison likely to cause death d) The therapeutic dose of a poison

4. Which parameter indicates whether sufficient oxygen is present for complete combustion?

   a) Flash point b) Vapour pressure c) Oxygen balance d)Ignition temperature

5. Pipe bombs are examples of:

   a) Military explosives b) Improvised Explosive Devices c) Commercial explosives

   d) Industrial explosives

6. The Biological Passport is mainly used in:
   a) Crime investigation b) Anti-doping programmes c) Drug trafficking d)Toxicology
7. The interval between administration of a poison and death is called the:
   a) Incubation period b) Fatal period c) Latent period d)Elimination period
8. Which agency primarily enforces the NDPS Act?
   a) WHO b) NCB c) UNESCO d) RBI
9. Which alcohol causes permanent blindness?
   a) Ethanol b) Methanol c) Propanol d) Butanol
10. Which of the following is the most common basis for the classification of poisons?
    a) Colour of the poison b) Taste of the poison c) Mode of action on the body
    d) Cost of the poison
11. Which specimen is most suitable for detecting recently ingested poison?
    a) Bone marrow b) Stomach wash/vomitus c) Fingernails d) Teeth

[1x11=11]

Part B

Short Answer

Answer any **8** questions

Each question carries **3** marks

12. What are liquid incendiaries? Give examples.
13. Mention four environmental effects of explosions.
14. Define controlled drugs and controlled substances.
15. What are designer drugs? Mention four examples.
16. Differentiate alcoholic and non-alcoholic beverages.
17. Classify poisons based on their mode of action, giving one example for each class.
18. What are antidotes? Classify antidotes and give one example of each type.
19. Write a short note on the post-mortem findings in cases of poisoning.
20. Describe the common signs and symptoms of corrosive poisoning.
21. Why is interpretation of toxicological findings important in forensic investigations?

[3x8=24]

Part C

Short Essay

Answer any **3** questions

Each question carries **5** marks

22. Discuss the effects of explosion on the human body, buildings.
23. Explain designer drugs. Discuss their classification.
24. Classify poisons based on their mode of action, giving one example for each class.
25. What are antidotes? Classify antidotes and give one example of each type.
26. Write a short note on the post-mortem findings in cases of poisoning.

[5x3=15]

Part D

Essay

Answer any **2** questions

Each question carries **10** marks

27. Explain the mechanism of explosion with suitable diagrams.
28. Describe the post-mortem findings commonly observed in cases of poisoning. Explain their forensic significance.
29. Describe the preparation of a toxicological report and discuss its essential components.

[10x2=20]